

Is there a general formula for the ratio limit of $a(n) = a(n-1) + a(n-m)$.?

Asked 2 years, 10 months ago Modified 2 years, 10 months ago Viewed 82 times



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I am exploring the properties of a generalization of Fibonacci numbers and have been encountered an issue.

Specifically I am trying to find the limit of the ratio of the the sequence:

$$A(n) = A(n-1) + A(n-m)$$

$$A(n) = n+1 \text{ for } 0 \leq k < m$$

when $m = 2$ this is the Fibonacci sequence.

It's clear that the solution is the real root of the polynomial:

$$x^m - x^{(m-1)} - 1 = 0$$

For $n=1$ the ratio is 2, $n=2$ is the golden ratio and $n=3$ is the supergolden ratio. But beyond that I haven't made much progress. I have written a script to generate the sequences and approximate the ratio which has given the following for the for $1 \leq m \leq 10$:

```
1: 2.0: [1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096, 8192, 16384, 32768, 65536, 131072, 262144, 524288]
2: 1.618033988749895: [1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987, 1597, 2584, 4181, 6765, 10946]
3: 1.465571231876768: [1, 2, 3, 4, 6, 9, 13, 19, 28, 41, 60, 88, 129, 189, 277, 406, 595, 872, 1278, 1873]
4: 1.3802775690976141: [1, 2, 3, 4, 5, 7, 10, 14, 19, 26, 36, 50, 69, 95, 131, 181, 250, 345, 476, 657]
5: 1.324717957244746: [1, 2, 3, 4, 5, 6, 8, 11, 15, 20, 26, 34, 45, 60, 80, 106, 140, 185, 245, 325]
6: 1.2851990332453493: [1, 2, 3, 4, 5, 6, 7, 9, 12, 16, 21, 27, 34, 43, 55, 71, 92, 119, 153, 196]
7: 1.2554228710768465: [1, 2, 3, 4, 5, 6, 7, 8, 10, 13, 17, 22, 28, 35, 43, 53, 66, 83, 105, 133]
8: 1.2320546314285723: [1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 14, 18, 23, 29, 36, 44, 53, 64, 78, 96]
9: 1.21314972305964: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 15, 19, 24, 30, 37, 45, 54, 64, 76]
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Is there a known solution for this? Or a proof there isn't one?

Thank you

[recurrence-relations](#) [fibonacci-numbers](#) [ratio](#)

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asked Mar 3, 2021 at 21:56

 **Shahar**
Shahar 1

2  What do you regard as a *solution*? You have a recurrence and a generating function is not difficult. The ratio limit is precisely the real root of $x^n - x^{m-1} - 1 = 0$ precisely as you have stated – [Henry](#) Mar 3, 2021 at 22:02 

 You're right. I should've been more precise. The generating function is $f(x) = 1/(1-x-x^m)$. Would it be possible to represent the roots using continued fractions? – [Shahar](#) Mar 3, 2021 at 22:11 

2 Answers

Sorted by: [Highest score \(default\)](#) 



The sequence that you are looking at is similar to the Fibonacci-Narayana sequence, which is given by

$f_n = f_{n-1} + f_{n-1-m}, m \geq 1$. The generating function is given by

1

$$\frac{1}{1 - x - x^{n+1}}$$

and the limiting ratio is given by

$$\lim_{n \rightarrow \infty} \frac{f_{n+1}}{f_n} = \chi$$

$$\chi - 1 = \chi^{-n}$$

This sequence can be found in OEIS A000930 and related. $n = 1, 2, 4$ are the golden ratio, supergolden ratio, and the plastic number, respectively.

Additional properties of this sequence are

$$\sum_{n=0}^{\infty} \frac{1}{\chi^n} = \chi^{n+1}$$

$$S_n = \sum_{j=1}^n f_j \quad (\text{cumulative sum})$$

$$\lim_{n \rightarrow \infty} \frac{S_n}{S_{n+1}} = \chi$$

You might also be interested in the Fibonacci-Padovan series, $f_n = f_{n-m+1} + f_{n-m}$ with a limiting value given by $X + 1 = X^m$.

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edited Mar 4, 2021 at 21:43

answered Mar 4, 2021 at 21:37



Cyé Waldman

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I am not sure what you want. You have essentially specified the answer to be a root of a certain polynomial equation.

If you want to express the solution in terms of radicals (i.e. n^{th} roots) like $\phi = \frac{1+\sqrt{5}}{2}$ then I am afraid this is impossible, and the reason has to do with Galois theory, so you can ignore the following if you are unfamiliar with Galois theory.

But if you are, then e.g. the case where $m = 7$ we have the polynomial $x^7 - x^6 - 1$ whose Galois group is S_7 (computed with PARI/GP) which is not solvable, so its roots cannot be expressed using radicals.

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answered Mar 3, 2021 at 22:01



Kenny Lau

25k 31 74



I am familiar with Galois theory and due to the fact this polynomial has a degree larger than 5, a general solution is not possible.



I am looking for other solutions. – [Shahar](#) Mar 3, 2021 at 22:09 ✎